
Mathematics Competition

Student Exchange Germany – China

– Task Sheet –

Name: _____

Tips received: ☐ Task 1 ☐ Task 2

Welcome to the competition!

We are glad you are here! Today is all about the joy of problem-solving, logical thinking, and the great feeling of cracking a tricky puzzle (the famous "Aha! moment").

The Rules:

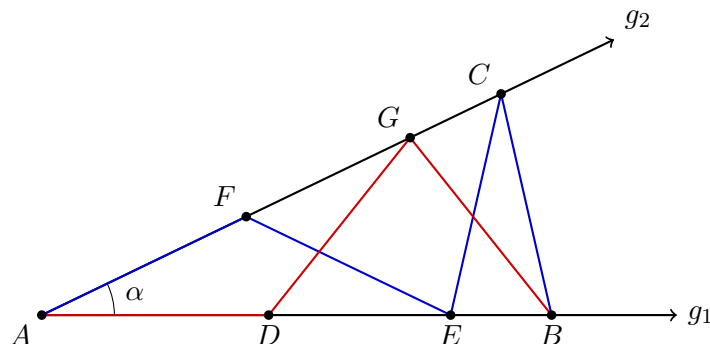
- **Time limit:** You have exactly **30 minutes** for the two tasks on the next page.
 - **Tip cards:** If you get stuck, you can get some help:
 - After **10 minutes**, you can ask for exactly **one tip** per task.
 - **Scoring:** **Only the correct result written in the designated box counts** (no derivations or justifications are required).
 - **2 points** for the correct result in Task 1.
 - **1 point** for each correct result in Task 2 (max. 2 points).
 - **Minus 0.5 points** penalty for each tip used!
 - **Who wins?** The winners are determined solely by the final score. Whoever has the most points at the end wins. There can be multiple winners!
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Task 1: The Seven-Segment Figure

Given are two rays g_1 and g_2 , intersecting at a point A with an acute angle α . We construct a geometric zigzag line between these rays. For this, we choose points D, E, B on g_1 and F, G, C on g_2 such that the following **seven segments all have the exact same length**:

$$|\overline{AD}| = |\overline{AF}| = |\overline{FE}| = |\overline{DG}| = |\overline{EC}| = |\overline{GB}| = |\overline{CB}|$$

The final segment $\overline{CB}$ perfectly closes the figure (see sketch).



Question: How large must the starting angle α be so that this zigzag figure can close after exactly seven segments of equal length? Enter your result into the box.

$\alpha =$

Task 2: Large Products – Many Zeros?

a) With how many zeros does the following product end?

$$1 \cdot 2 \cdot 3 \cdots 99 \cdot 100 ?$$

Number of trailing zeros:

b) Determine the smallest natural number n for which the product

$$1 \cdot 2 \cdot 3 \cdots n$$

ends with exactly 250 zeros.

$n =$